Summary of Data-Institution Context: Flemish Educational System

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1 Flemish Educational Context

Education in Flanders is compulsory from ages 6 to 18 and almost entirely publicly funded. Around 99% of pupils attend subsidized schools—with no tuition fees and only modest parental contributions (typically €300–400 per year for materials or excursions).

Primary education (ages 6–12) is free and open-access: families choose among nearby schools, with enrolment coordinated locally rather than through competitive entry.

Secondary education begins at age 12 and remains publicly funded. Students start in a common first stage before being oriented into one of several tracks: general (ASO), technical (TSO), vocational (BSO), or arts (KSO). This early tracking defines later educational and career opportunities; movement between tracks is possible but limited.

Higher education—universities, university colleges, and professional institutes—is also publicly subsidized. Tuition fees are low (around €1,000 per year). Access is generally open, except for some studies (e.g, medical studies). Because access costs are low and travel distances short, study choices largely reflect student preferences rather than financial or geographical constraints. The first year, however, serves as an informal selection stage, with high dropout and switching rates (Declercq and Verboven, 2015, 2018).

1.1 Secondary Education: Students

Stages of Secondary Education

Stage	**Age**	**Description**
1st stage	12–14	Orientation stage. Students enter the A-stream (general) or B-stream (remedial/practical)
2nd stage	14–16	Choice of study type (*onderwijsvorm*): ASO, TSO, BSO, or KSO.
3rd stage	16–18	Further specialization within the chosen track leading to graduation.
Optional 7th year	18–19	For BSO students to obtain a full diploma and access higher education.

See Table 1 in the appendix for a schematic overview.

Study Types (Tracks)

- **ASO**: General secondary education – theoretical and academic; prepares for university.
- **TSO**: Technical secondary education – combines theory and practice; leads to higher professional education or work.

- **BSO**: Vocational secondary education – practice-oriented; leads to work unless a 7th specialization year is completed.
- **KSO**: Arts secondary education – artistic and general formation; prepares for specialized higher education.
- Within ASO and TSO, students also specialize by subject intensity (STEM vs. non-STEM, humanities, etc.). These subtrack choices are highly predictive of later higher-education fields, as shown in the LOSO data (see Figure 2).

Certificates and Restrictions on Progression

- **A-certificate**: Full pass; student can progress to any track.
- **B-certificate**: Conditional pass; student must move to a lower track or repeat[1].
- **C-certificate**: Fail; student must repeat the year.
- These decisions can also occur within subtracks (e.g. a student may be required to change the math intensity level within ASO).

Post-secondary Transitions

- ASO graduates typically enter university or college.
- TSO and KSO students often pursue higher professional education.
- BSO graduates usually enter the labor market unless they complete the optional 7th specialization year.
- Subtrack choices strongly predict the type of higher education pursued (see Figure 2).

School Types and Costs

- Around 99% of Flemish secondary students attend publicly funded schools; education is tuition-free, with only small parental contributions.
- Some schools offer only specific tracks, creating sorting patterns across "elite" and mixed schools (DeGroote and Declercq, 2018).
- Teacher wages are centrally fixed, so teachers tend to sort based on non-wage amenities (school composition, proximity, or working climate).

Entry to Higher Education

- Universities and colleges are low-cost and mostly open-access.
- Admission exams apply only to medicine, dentistry, and veterinary programs; STEM programs require a non-binding start test.
- Choices at the end of secondary school thus reflect preferences more than constraints, though the first year acts as a de facto selection mechanism (Declercq and Verboven, 2018).

1.2 Secondary: Teachers and Schools

Contracts and Stability

- Teachers start on temporary contracts, gaining priority rights for reappointment.
- After several years, they can obtain tenure, a civil-service–like permanent status.
- Tenure provides strong job security: surplus tenured staff are reallocated rather than dismissed[2]

[1] studied in detail in (De Groote, 2025)

[2] Information on teachers is based on OECD (2023) *Education at a Glance: Belgium Country Note* and Flemish Ministry of Education administrative rules (AGODI).

Training and Entry Requirements

- **Primary:** requires a three-year professional bachelor in education.
- **Secondary:** requires either an educational bachelor (for lower secondary; e.g grades 1-2) or a master's degree plus teacher training (for middle and upper secondary, e.g grades 3-8).
- The qualification level (bachelor vs. master) determines pay scales and qualification type (STEM degree, humanities etc) subject eligibility.

Skill Heterogeneity

- Qualification level varies across schools and subjects.
- Persistent shortages exist in STEM, technical, and practical fields.
- Disadvantaged schools face higher turnover and attract fewer experienced teachers.
- Pay is standardized by qualification and seniority, not by performance; sorting occurs through non-wage school characteristics.

Outside Options

- Teacher salaries are high in absolute terms but less competitive relative to similarly educated workers in other sectors.
- The gap is largest for master-level teachers, especially in STEM, where outside options are stronger.
- Teaching offers stability, automatic wage indexation, and low exit risk, but career ladders are flat.

Hiring and Autonomy

- Teachers are employed by local school boards rather than directly by the Ministry.
- Boards have autonomy in hiring and evaluation within centrally defined pay and qualification frameworks.
- Thus, hiring is decentralized, but wages and contract types are uniform across the Flemish system.

Payment

Teacher Pay Structure (Flanders)

Payment depends on **training** and **type of subject**, not explicitly on the **track** (ASO, TSO, KSO, BSO). The classification below follows the official system, which maps the education required to the period of highschool and assigns a salary level (301, 501):

soort	**1st grade**	**2nd Grade ASO/TSO/KSO**	**3d grade ASO/TSO/KSO**	**2nd grade B**
Typical "academic" subjects				
Bachelor	301	301	...	301
Master	...	501	501	...
Example: Mathematics (same for Latin, Chemistry, etc.)				
More practical subjects (Mechanica, Bouwkunde, etc. – only offered in TSO/KSO)				
Bachelor / practical orientation	301	301	302	301
Master / theoretical training	...	501	501	...
Example: Mechanica, Bouwkunde				
Low-level practical subjects (e.g. Haartooi)				
301	...	...	...	...

Takeaway. Wages differ *not by track*, but because tracks contain different types of subjects and thus attract teachers with different qualification levels. A school that offers ASO–TSO–BSO can employ master-qualified teachers (501) teaching Dutch or Math in all tracks — earning the same across them. However, teachers in TSO or BSO who teach more practical subjects fall under lower qualification categories (301–302).

Scale	**Qualification**	**Min salary (€/year)**	**Max salary (€/year)**	**Typical teacher**
301	Bachelor / Educatief graduaat	17,538	31,485	Leraar–regent (richtgraad)
501	Master / Educatieve master	21,965	40,064	Leraar–licentiaat geaggregeerde

Scale difference (valid from 01/09/2025).

2 Data

2.1 LOSO 1991-1997

Starting in September 1990, a cohort of **6,400 students** was tracked throughout their educational careers in **89 secondary schools**. These schools were selected from two regions: **Dendermonde-Zele-Hamme** and **Diest-Hasselt-Genk-Houthalen-Leopoldsburg-Beringen**, based on criteria such as school size, administrative structure, educational type, and representation of different school networks (public and private education).

Several types of surveys were conducted to accommodate the study's complex design, including:

- Continuous tracking of students' educational pathways at the start and end of each school year.
- Baseline assessment at the beginning of secondary school, including intelligence tests, motivation, academic performance, school choice, and teacher evaluations.
- Evaluations at the end of the **first, second, and fourth years**, measuring school performance, self-concept, well-being, interest, and career choices.
- Parent surveys in the first year.
- Follow-up surveys in the sixth year and for students who had repeated a grade.
- Assessment of school characteristics through surveys with principals, teachers, and students.
- Post-secondary career tracking for at least **3.5 years** after graduation.

2.2 Students' surveys

Table 1: Student-level data collection

	b. 90/91	**e. 90/91**	**91/92**	**92/93**	**93/94**	**94/95**	**b. 95/96**	**e. 95/96**
Study career data	x		x		x		x	
Intelligence	x			x				
Performance motivation	x							
Parent questionnaire	x							
Dutch test	x	x		x		x		x
Math test	x	x		x		x		x
Well-being/self-concept		x		x		x		x
Choice questionnaire	x	x		x		x		x
Interests		x		x			x	
BC questionnaire (norm.)				x		x		
Survey delayed students				x		x		

2.3 Teachers' surveys

Table 2: Teacher-level data collection

	b. 90/91	e. 90/91	91/92	92/93	93/94	94/95	b. 95/96	e. 95/96
Teacher assessment	x	x		x		x		x
Choice advice (teachers)	x	x		x		x		x
Opportunity to learn		x		x		x		x
Teaching practices			x		x		x	
Class evaluation			x		x		x	
Study choice guidance		x		x				

2.4 Schools's surveys

Table 3: School-level data collection

	b. 90/91	e. 90/91	91/92	92/93	93/94	94/95	b. 95/96	e. 95/96
School char. (directors)		x						
School char. (teachers)		x						
School char. (students)			x		x		x	

* Student surveys were also conducted one year later with pupils who experienced a study delay.

2.5 Overview of Surveys

Student-level Surveys

- **Parental Questionnaire (Start 1st Year)** – family composition, educational climate, school-/education orientation of the family, school-/study choice, school history of the child, personality of the child, education level of the family, occupational situation, economic-cultural-social milieu.
- **Intelligence Tests** –
  - Start 1st Year: verbal (synonyms, antonyms, word categories), numerical (number sequences, arithmetic), spatial (cut figures, figure categories) (GETLOV).
  - Start 6th Year: Berenschot G-test.
- **Performance Motivation (Start 1st Year)** – performance motivation, positive test anxiety, negative test anxiety, social desirability (motivation also returns as a theme in the well-being/self-concept questionnaire).
- **Dutch Tests** –
  - Start 1st Year: spelling, grammar, idiomatic language, silent reading.
  - End 1st Year A: spelling, sentence reading, information processing, language use, silent reading.
  - End 1st Year B: spelling, grammar, information processing, silent reading.
  - End 2nd Year GL: spelling, grammar, language use, silent reading.
  - End 2nd Year BVL: spelling, grammar, information processing, silent reading.
  - End 4th Year: level-based tests.
  - End 6th Year: level-based tests.
- **Mathematics Tests** – analogous to Dutch; separate versions per grade; distinction A-B and GL-BVL; recurring parts: number theory and geometry. Note: in the 2nd Year GL specific questions for each of the two school networks. End 4th Year: level-based tests. End 6th Year: level-based tests (no test in BSO).

- **Well-being / Self-concept (End 1st, 2nd, 4th, and 6th Year)** – integration in the class, relationship with teachers, feeling good at school, concentration in class, homework attitude, interest in learning tasks, academic self-concept, effort in learning tasks.
- **Choice Questionnaire** –
  - Start 1st Year: primary school experiences, school choice, determinants of study choice, coping mechanisms, future perspective.
  - End 1st + 2nd Year: determinants of study choice, coping mechanisms, future perspective, provisional study choice.
  - End 4th Year: shortened version of 2nd Year questionnaire, plus exploration of career perspective.
  - End 6th Year: analogous to End 4th Year.
- **Interest Questionnaire** –
  - End 1st Year: OII – personal/social service, outdoor activities, technology, commerce, arts, sciences, literature, verbal type, manipulative type, numeric type.
  - End 2nd Year: BSTR – economy/commerce, fine technology, health care, clothing/fashion, agriculture, physical education/sport, hospitality/service, practical technical, beauty care, decorative arts, social service, languages, theoretical technical, nutrition/food preparation, sciences, mathematics.
  - Start 6th Year: OII – as at End 1st Year.
- **B-/C-Certificate Questionnaire (End 3rd and 5th Year)** – experience of receiving this certificate, causes, decisions afterwards and reasons, subsequent school choice, future perspective, impact of the certificate and its subjective importance.
- **LOSO Annex Questionnaire (3.5 years after 6th Year or leaving secondary)** – family situation, important life events, future perspective; higher education (registration data, experience of transition, institution choice); labour market (registration of periods of unemployment/jobs/training, recruitment channel, function/tasks of first and current/last job, occupational category, monthly net salary, unemployed status, reason for not working).

Teacher-level Surveys

- **Teacher Assessments of Students** –
  - Start 1st Year: global personality assessment, evaluation of striking behaviours, school career during primary school, study results: Dutch–Mathematics–total.
  - End 1st + 2nd Year: global personality assessment, study results (Dutch–Mathematics + interest/effort), total: placement in group.
  - End 4th Year: global personality assessment, study results (Dutch–Mathematics + interest/effort), assessment by two subject-specific teachers, total: placement in group.
  - End 6th Year: analogous to End 4th Year.
- **Choice Advice by Teachers** –
  - Start 1st Year: advisory behaviour, vision short + long term.
  - End 1st Year: short-term vision (next year).
  - End 2nd Year: short- and long-term vision.
  - End 4th and 6th Year: repetition of End 2nd Year.
- **Opportunity to Learn (End 1st, 2nd, 4th, 6th Year)** – survey of Dutch and Mathematics teachers: which questions from the tests belonged to the taught (and required) material.
- **Teaching Methods (End 2nd, 4th, 6th Year)** – survey of Dutch and Mathematics teachers: concrete classroom practice (textbooks, lesson content, classroom events), pupil response, expectations of pupils, satisfaction, collaboration between teachers.

- **Classroom Assessment (End 2nd, 4th, 6th Year)** – survey of Dutch and Mathematics teachers: class group cohesion, participation, task orientation and learning orientation, competition, order and organization, satisfaction with capacities.
- **Study Choice Guidance (End 1st + 2nd Year)** – survey of the (formal) method of study choice guidance by homeroom teachers (klastitularissen).
- **Teacher Survey (End 1st Year)** – personal background, work experience, lesson preparation, handling of problem behaviour, time use, daily teaching practice, tests and exams, educational goals, school/educational orientation, cooperation structures, decision-making, internal school relations, relations with external actors, regulations, school as organization, deliberations and orientation.
- **Administrative Teacher Data (1990–91)** – age, gender, diploma, employment status, number of years of experience, number of subjects taught, number of teaching hours in different grades.

School-level Surveys

- **School Management Survey (End 1st Year)** –
  - general characteristics: current structure and recent changes, facilities, administrative data,
  - personal background of the principal, time allocation and pedagogical tasks (also of vice-principals),
  - policy and organizational characteristics: timetable, appointments, workgroups, class composition, pupil evaluation, pupil guidance, decision-making, consultative bodies, relations with external actors (e.g. PMS, other schools), satisfaction.
- **School Characteristics – Student Questionnaire (End 1st, 3rd, and 5th Year)** – participation and independence of pupils; learning orientation and task orientation of pupils and teachers; relationships among pupils and with teachers; order and organization, clarity of rules; guidance, study choice process; differentiation, competition, innovation, activities and facilities; satisfaction. (Same version for all pupils, including normal-progress and sample groups.)

2.6 Data Storage

All data are stored in 21 **ACCESS 2000** databases. This includes not only data from the initial sample, called the "LOSO sample group," but also from students who, at a certain moment or for a certain period, were in the same class as the LOSO sample group.

Student Data

The **educational trajectories** of students are stored in the database `loopbanen.mdb`. Trajectory data (school, grade, year, educational track, study orientation, curriculum, certificate) were continuously registered for all 6,400 sample students from 1989–90 (last year of primary education) until the beginning of 1997–98, as long as the student was still in some form of secondary education.

For non-sample students, registration of trajectory data was limited to the school years in which they participated in classroom-level surveys. Data on the transition to higher education and/or the labor market were registered for both sample and non-sample students, using information from the follow-up questionnaire and from universities, colleges, and the VDAB (employment office).

Note: Educational trajectories from primary education are stored in `basisonderwijs.mdb`.

Nr.	Instrument	Database
1	Parental questionnaire	Oudervragenlijst.mdb
2	Intelligence tests	Intelligentie.mdb
3	Performance motivation	PMTK.mdb
4	Dutch tests	Schoolvorderingentoetsen.mdb
5	Mathematics tests	Schoolvorderingentoetsen.mdb
6	Well-being / Self-concept questionnaire	Eindvragenlijsten.mdb
7	Choice questionnaire	Keuzevragenlijsten.mdb
8	Interest questionnaire	Belangstelling.mdb
9	Student school climate questionnaire	Schoolkenmerken_leerlingen.mdb
10	B-/C-certificate questionnaire	Vragenlijsten_bc-attest.mdb
11	Social variables	Sociale_variabelen.mdb
12	Follow-up questionnaire	Vragenlijst_follow-up.mdb

Table 4: Student instruments and associated databases

Student Instrument Data

Teacher and School Data

Nr.	Instrument	Database
11	Teacher assessment (beg 1st year)	Basisonderwijs.mdb
12	Teacher choice advice (beg 1st year)	Basisonderwijs.mdb
13	Teacher assessment (end 1st year)	Leerkrachten_graad_1.mdb
14	Teacher assessment (end 2nd year, incl. 2bis)	Leerkrachten_graad_1.mdb
15	Teacher choice advice (end 1st year)	Leerkrachten_graad_1.mdb
16	Teacher choice advice (end 2nd year, incl. 2bis)	Leerkrachten_graad_1.mdb
17	Opportunity to learn (end 1st year)	Leerkrachten_graad_1.mdb
18	Opportunity to learn (end 2nd year)	Leerkrachten_graad_1.mdb
19	Teaching methods (end 2nd year)	Leerkrachten_graad_1.mdb
20	Classroom evaluation (end 2nd year)	Leerkrachten_graad_1.mdb
21	Study choice guidance (end 1st year)	Leerkrachten_graad_1.mdb
22	Study choice guidance (end 2nd year)	Leerkrachten_graad_1.mdb
23	Teacher assessment (end 4th year)	Leerkrachten_graad_2_93-94.mdb
24	Teacher assessment (end 4th year, bis)	Leerkrachten_graad_2_94-95.mdb
25	Teacher choice advice (end 4th year)	Leerkrachten_graad_2_93-94.mdb
26	Teacher choice advice (end 4th year, bis)	Leerkrachten_graad_2_94-95.mdb
27	Opportunity to learn (end 4th year)	Leerkrachten_graad_2_93-94.mdb
28	Teaching methods (end 4th year)	Leerkrachten_graad_2_93-94.mdb
29	Classroom evaluation (end 4th year)	Leerkrachten_graad_2_93-94.mdb
30	Teacher assessment (end 6th year)	Leerkrachten_graad_3_95-96.mdb
31	Teacher assessment (end 6th year, bis)	Leerkrachten_graad_3_96-97.mdb
32	Teacher choice advice (end 6th year)	Leerkrachten_graad_3_95-96.mdb
33	Teacher choice advice (end 6th year, bis)	Leerkrachten_graad_3_96-97.mdb
34	Opportunity to learn (end 6th year)	Leerkrachten_graad_3_95-96.mdb
35	Opportunity to learn (end 6th year, bis)	Leerkrachten_graad_3_96-97.mdb
36	Teaching methods (end 6th year)	Leerkrachten_graad_3_95-96.mdb
37	Classroom evaluation (end 6th year)	Leerkrachten_graad_3_95-

2.7 Tables

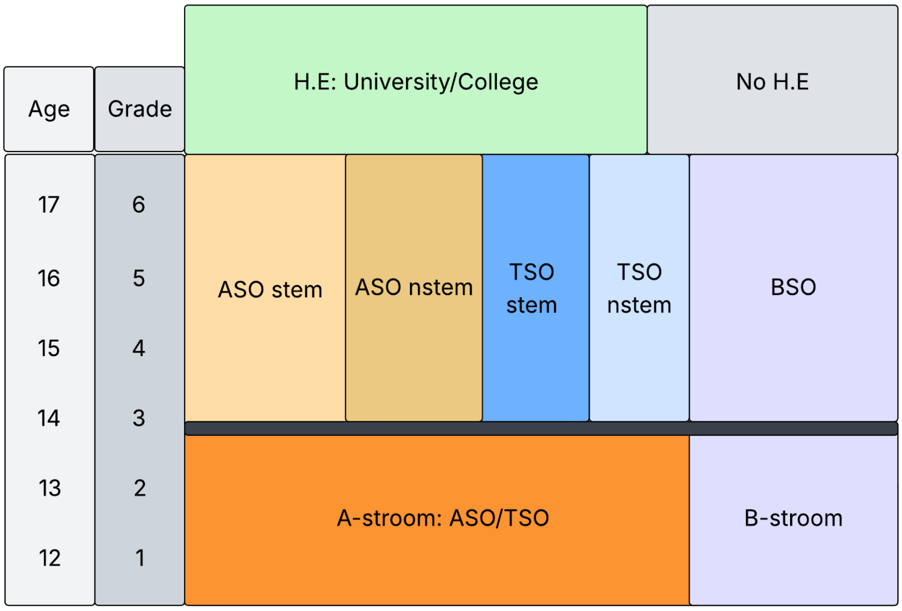

Figure 1: Overview of the Belgian track system in high school. The first two years are general, but at age 14 students must make a binding choice between academic (ASO), technical (TSO), or vocational (BSO) tracks. They must also choose the degree of STEM or noSTEM education they want to receive in each track. This paper focuses on this key transition period, marked in black, when students move from general to specialized education and where tracking effectively starts.

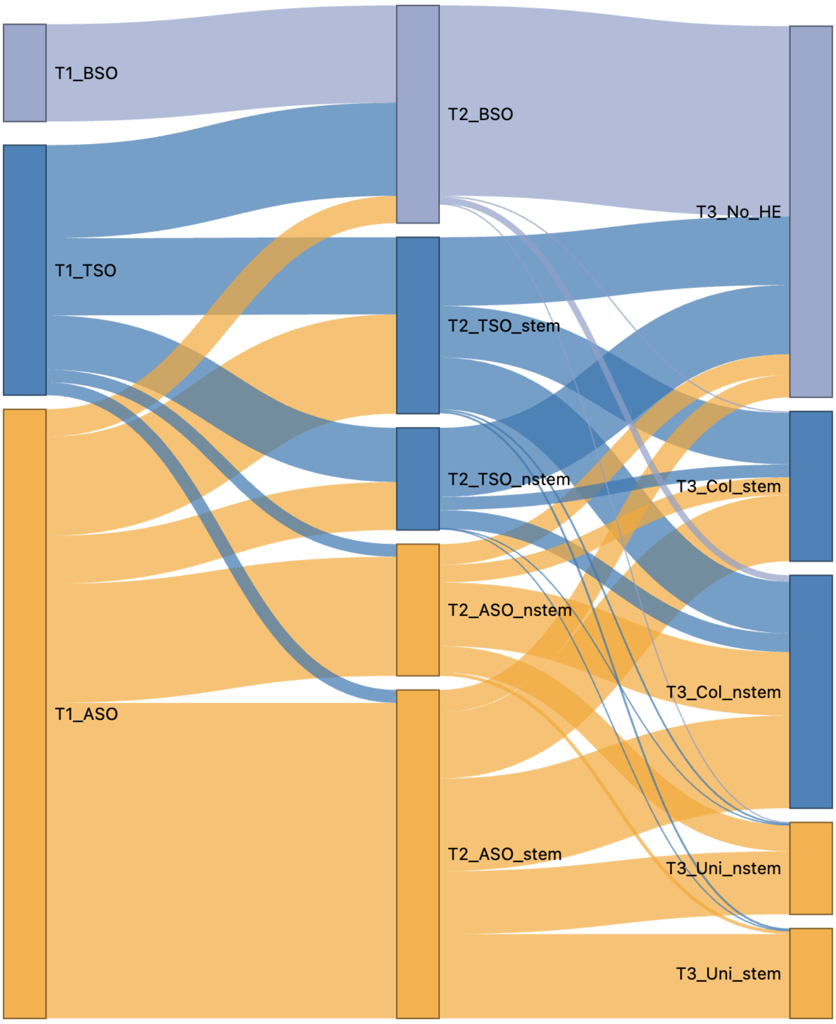

Figure 2: Student track and subtrack transitions across educational stages: grade 1 (t_1), grade 3 (t_2), and higher education entry (t_3). The graph is based on the restricted sample at t_2, linking each student's past (track at grade 1) and future (field of study at higher education). Transitions are color-coded to highlight the direction and magnitude of flows across tracks. The figure was generated using the `networkD3` package in R

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